

When Is a Ground-Truth-Free Reliability Signal Real? Difficulty Confounding and Its Scale Dependence in Multi-Agent LLM Consensus

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Abstract

Multi-agent LLM systems often *converge on wrong answers*—false consensus—and a natural hope is that cheap, ground-truth-free signals (inter-agent answer agreement, self-reported confidence) can flag these failures online. We show why the naive evaluation cannot establish this claim. On a single MATH Level-5 slice, round-2 agreement is moderately predictive ($|AUC| \approx 0.68$ on a balanced high-confidence subset) but falls to 0.51 after adjustment for the only available within-slice difficulty proxy, the backbone’s own solo error rate. Because that proxy is circular, the result neither validates nor cleanly refutes the signal; it exposes an identification failure. We then construct non-circular cross-level controls and ask what survives them. We introduce a four-gate audit protocol—within-stratum discrimination, collider-bias, independent-model residual difficulty, and non-circular difficulty strength—that tests whether a signal’s predictive power survives explicit difficulty controls, and we apply it across four model scales (Qwen2.5 3B/7B/14B/32B) on MATH with matched item pools and machine-verified answers ($n \approx 200$ each). **(i) In the Qwen–MATH series, answer-agreement’s predictive power survives every specified control at all four scales**, with a within-stratum AUC of 0.72–0.80 in the archived MATH traces across an order of magnitude in model size; fresh 3B/7B/14B repeats are 0.70–0.77. **(ii) Confidence conclusions depend on which measure is reported.** Its mean true-vs-false separation dips at 14B and recovers at 32B; the dip is significant on the bounded probability scale but not detectable on rank AUC or a variance-standardised gap. Those separations are pooled; within difficulty stratum the 3B value is in fact *negative*, so even the sign of the small-scale effect depends on the estimand. We note why these can diverge—rank functionals are invariant under any monotone reparameterisation of the confidence scale, mean gaps are not—so a shrinking gap is evidence of lost discrimination only if it survives a rank-invariant measure. Reporting a single measure here would have produced a confident and wrong scaling story, in either direction. **(iii) The agreement signal has a candidate domain boundary.** On a disjoint MMLU holdout it is substantially weaker across five models at or below 14B (AUC = 0.54–0.64; fresh repeats 0.52–0.65, versus their MATH runs); the earlier 32B stress test was 0.58. A simple sampling model offers a diagnostic account: under a one-preferred-answer approximation, agreement is informative only through the difference in *top mass* between correct and incorrect consensus, and the MMLU fit implies a wrong answer still carries most of that mass, so three agents repeat the same error. This is consistent with agreement detecting idiosyncratic reasoning slips rather than shared knowledge gaps. We release the audit protocol, the pools, and all runs.

1 Introduction

Multi-agent large language model (LLM) systems—debate, self-consistency, and role-based collaboration—are increasingly deployed in the hope that several models will catch each other’s mistakes. In practice they frequently do the opposite: they *converge on a wrong answer*, a failure mode that

recent work calls *false consensus* or premature agreement [Cemri et al., 2025, Wu et al., 2025]. Diagnosing this failure today is largely an *offline*, post-hoc exercise requiring ground-truth labels or human annotation. A practically appealing alternative is an *online*, ground-truth-free signal: if inter-agent answer agreement or the agents’ self-reported confidence could tell true consensus from false consensus *during* inference, one could gate, escalate, or repair the collaboration at almost no cost.

This paper is a cautionary tale and a positive result at once. We began by testing the naive claim on a single, hard slice of MATH (Level 5) and found what looked like a usable detector: round-2 answer-agreement separated true from false consensus with $|AUC| \approx 0.68$ on a balanced, high-confidence subset. Adjusting it for the only available within-slice difficulty proxy—the backbone’s own solo error rate—reduced it to $|AUC| \approx 0.51$. That adjustment is itself circular, so it is not a clean difficulty control and does not establish that agreement is an artifact. Instead, it demonstrates that the single-slice design cannot identify the claim. The reason is structural: harder problems both (a) make the agents more likely to disagree or hedge, and (b) make them more likely to be wrong. Any signal that merely tracks difficulty will look predictive of error while carrying no information about *consensus quality* beyond difficulty itself. This is a difficulty-confounding trap, and we show it is easy to fall into and easy to mistake for a real finding.

The trap, however, is not the end of the story—it is a measurement problem. The single-slice design removed the very axis (difficulty) needed to test the claim, because within one official difficulty band the only usable difficulty estimate is the model’s own error rate, which is *circular*. Our first contribution is a diagnosis and a fix: build a *cross-level* item pool (MATH Levels 1–5) so that official human difficulty becomes an independent, high-variance control, and then subject every candidate signal to a **four-gate audit**:

1. **Within-stratum discrimination.** Does the signal separate true/false consensus *inside* a fixed official difficulty band (removing between-band differences while acknowledging within-band variation)?
2. **Collider bias.** Is the within-stratum signal an artifact of conditioning on convergence, or is it already present on the full item set?
3. **Residual difficulty.** Does the signal survive after removing a *fine-grained*, non-circular difficulty estimate from an *independent* model family (Mistral-based Mathstral)?
4. **Non-circular control strength.** Is the difficulty control we used actually strong (monotone false-consensus rate, high $|AUC|$), so that “survives the control” is a meaningful statement?

Our second contribution is to ask whether a signal that passes this audit is *stable across model scale*, or an accident of one backbone. We run the identical protocol—same pool, same debate, same machine-verified answers—on four sizes of the same family (Qwen2.5 3B, 7B, 14B, 32B; $n \approx 200$ items each), which span an order of magnitude:

- **Answer-agreement’s predictive power survives every control we ran.** At all four scales its within-stratum AUC stays in 0.72–0.80, varying by 0.073 across an order of magnitude in model size. This is the signal worth building on, subject to the caveats in Section 8 about what our controls can and cannot rule out.
- **Confidence conclusions depend on which measure is reported.** Its mean true-vs-false separation runs +0.014 (3B), +0.049 (7B), +0.024 (14B), +0.057 (32B)—a dip at 14B that recovers. The dip is significant on the bounded probability scale but not detectable on rank AUC or on a variance-standardised gap, and 14B’s confidence SD is about half of

7B’s. Compression is a plausible account; with n_F of 39–56 we cannot establish that the ordering is unchanged, only that we do not detect a change. The small-scale end is worse than “uncalibrated”: those separations are pooled, and *within* Level the 3B gap turns negative (−0.013 on the probability scale, −0.98 on the logit scale) with a rank AUC of 0.488, so the apparently positive pooled value at 3B is a between-Level artifact rather than weak signal.

This is where our own earlier reading went wrong, and the correction is the methodological point. With three points (3B, 7B, 14B) the separation looks like it peaks mid-scale, and we concluded that scaling claims need a third point. The fourth point reverses that reading—but more importantly, no number of points would have fixed it, because every point would have been measured on the same distorting scale. The underlying fact is elementary and not new (Section 6): rank functionals are invariant under *any* monotone reparameterisation of the confidence scale, mean gaps are not, so a model that merely grows more confident must show a shrinking gap and cannot show a changing AUC. Our contribution here is empirical—a concrete case where the two readings diverge materially—together with the resulting procedural recommendation.

Our third contribution is a candidate boundary. Answer-agreement has the same point-estimate pattern on ARC, although several bootstrap intervals cross our survival threshold; on **MMLU it is substantially weaker across multiple models**: the original 32B stress test is $\text{AUC} = 0.58$, and a disjoint holdout gives 0.54–0.64 across five models at or below 14B; a fresh repeat gives 0.52–0.65 in the same ordering. The differing MCQ protocol means these are domain stress tests, not direct causal comparisons with MATH. A simple sampling model suggests why: under a one-preferred-answer approximation, agreement is informative only through the difference in *top mass*—the probability of the model’s single most likely answer—between correct and incorrect consensus. Fitting that model to MMLU implies a wrong answer still carries most of the mass, so three agents reproduce the same error and unanimity is uninformative. This is consistent with agreement detecting idiosyncratic reasoning slips rather than shared knowledge gaps, and it points to a diagnostic a practitioner can run on a labelled audit sample.

Contributions. (1) We document the identification trap for ground-truth-free consensus-quality signals: on a single difficulty slice, the only available adjustment is a circular model-behavior proxy ($|\text{AUC}| : 0.683 \rightarrow 0.514$ for round-2 agreement), so a difficulty-independent claim is not testable there. (2) We give a reusable four-gate audit protocol, with a cross-level pool and an independent-model difficulty estimator, that tests whether a candidate signal’s predictive power survives explicit difficulty controls. (3) Across four scales we show that answer-agreement’s discrimination is stable under those controls, and—with a four-point, four-measure bootstrap—that single-measure scaling claims about confidence are under-determined regardless of how many scale points they use. (4) We document a domain on which the agreement signal fails, and give a sampling model that relates the failure to a measurable concentration property of the answer distribution. We release all code, pools, and logs.

2 Setup

Task and collaboration protocol. We study a minimal but representative multi-agent protocol: a two-round debate among three agents instantiated from a single backbone. Each item is a MATH problem [Hendrycks et al., 2021b] with a numeric answer. In round 1 every agent solves the problem independently and reports an answer $a_i^{(1)}$ and a self-assessed confidence $c_i^{(1)} \in [0, 1]$. Each agent then sees the others’ round-1 answers and may revise, producing $a_i^{(2)}, c_i^{(2)}$. We say the

system *converges* if a strict majority agrees on a single answer in round 2. A converged item is *true consensus* if the majority answer equals the gold answer and *false consensus* otherwise. All gold answers are machine-verified (numeric normalization); ambiguous items are dropped at pool-construction time. The “agents” are three stochastic samples from the same checkpoint and prompt, not independently trained or role-specialized models. Unless noted otherwise, generation uses Ollama with temperature 0.7 and a 1536-token cap. Seed 0 fixes item ordering, but generation itself is not seeded; the released JSONL files, rather than exact regeneration, are therefore the audit record.

Uncertainty convention. Unless otherwise stated, every reported bootstrap interval resamples *items* within its stated strata from one archived generation trace. It quantifies finite-item uncertainty conditional on those model outputs; it does *not* quantify variation across independent stochastic generations, because the Ollama backend did not expose a fixed sampling seed. We additionally make one fresh, independent generation repeat for each P0 configuration (Table 6) and report paired item-bootstrap trace differences. Two traces can check whether the point estimates materially change, but they still cannot estimate a generation-level population distribution. Accordingly, all inferential claims remain conditional on the released traces.

Signals. All signals are ground-truth-free—computable at inference time without labels. The two we focus on are:

- **Answer-agreement** r_1 -agree (and its round-2 analogue r_2 -agree): the fraction of agent pairs that agree on the answer.
- **Mean confidence** $\bar{c}$: the mean self-reported confidence across agents.

We also track reasoning-divergence (r_1 -div, r_2 -div; pairwise embedding distance of rationales), their round-over-round slopes, and confidence variance, giving eight candidate signals in total. The prediction target throughout is the binary label **false consensus** on converged items.

Cross-level pool. The pool contains 240 problems, 48 at each official MATH difficulty Level 1–5, balanced across four subjects (algebra, counting/probability, prealgebra, number theory). Difficulty labels are the dataset’s human labels, assigned *blind* to model behavior, so official Level is a non-circular difficulty axis with genuine variance—unlike a single-level slice, where the only within-slice difficulty estimate is the model’s own error rate.

Scales. We run the identical protocol and pool on four backbones from one family: Qwen2.5-3B, -7B, -14B, and -32B (instruct) [Team, 2025]. This holds the model family and protocol fixed while varying nominal parameter count by $10.7\times$; it does not isolate parameter count from every family-internal training difference. We collect $n \approx 200$ items per scale (exact converged counts in Table 2). To separate scale effects from family idiosyncrasies, we additionally run the same protocol and audit on two *independent* families—Llama-3.1-8B [Dubey et al., 2024] and Mistral-7B [Jiang et al., 2023]—and report the cross-family comparison in Section 5.

Execution record. The experiments ran on Windows with one NVIDIA RTX 3070 (8GB) using Ollama 0.33.3. For auditability, Appendix B records the exact local model weight SHA-256 identifiers, model tags, decoding settings, and artifact hashes. These identifiers describe the archived execution environment; they do not make unseeded generation bitwise reproducible.

Cross-domain stress tests. We additionally apply the same three-sample, two-round protocol to ARC [Clark et al., 2018], using its Easy/Challenge split as a two-stratum difficulty axis, for Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B (up to 240 items per run). At 32B, ARC yields too few false-consensus cases for interpretation, so the initial stress test used a 150-item MMLU pool [Hendrycks et al., 2021a]. We then construct a disjoint 200-item MMLU holdout, balanced between 100 STEM and 100 humanities/social-science questions, and run Qwen2.5-3B/7B/14B, Llama-3.1-8B, and Mistral-7B. The holdout uses a deliberately short, answer-and-confidence-only MCQ prompt (three samples, two rounds, 128-token cap), shared by all five models; it is therefore a matched cross-model domain stress test rather than a byte-for-byte repetition of the longer MATH protocol. The two-tier MMLU grouping is a coarse analysis proxy, not an official difficulty label; consequently all MMLU comparisons are domain stress tests rather than a full repetition of the MATH audit. ARC and MMLU use exact option-letter matching for gold evaluation.

Independent difficulty estimator. For the residual-difficulty gate we need a fine-grained, non-circular difficulty score. We use **Mathstral-7B** (a Mistral-family math model, independent of the Qwen backbones) and define D_{ext} as its empirical error rate on each item over repeated sampling. Being from a different family, D_{ext} cannot be circular with the Qwen agents’ own errors, and its item-level granularity is finer than the five official Levels ($|\text{AUC}|$ predicting Qwen error = 0.77–0.78; see Section 5).

3 The Four-Gate Audit

We ask a narrower question than “is this signal real”: whether a signal’s ability to predict false consensus *survives the specific difficulty controls we can construct*. We operationalize this with four gates. Passing them is evidence of robustness under those controls, not a certificate that the signal is free of difficulty; Section 8 states what they do and do not rule out. Throughout, separability is measured by the AUC of s against the false-consensus label.

Estimand and conventions. Two choices matter enough to state before any result. *First, the primary estimand is within-Level.* Because the whole point of the audit is to hold difficulty fixed, every AUC we use to support a difficulty-controlled claim—Table 2, Table 3, Table 4, and the left panel of Figure 2—is stratified by official Level. Pooled values are reported only where the quantity of interest is deliberately *not* difficulty-controlled (the operational gap a deployed monitor would read, and the sampling model of Section 6.2); those are labelled *pooled* wherever they appear. Table 1 shows both side by side precisely so the two can be compared. The within-Level estimate is *pair-weighted*: within each Level we score every (true, false) pair, and the reported value is the concordance summed over all pairs divided by the total pair count, so a Level contributes in proportion to $n_T^{(\ell)} n_F^{(\ell)}$. We use pair weighting rather than an equal-weight average of per-Level values because our strata are highly unbalanced—Level 1 has only 1–4 false-consensus items at every scale—and equal weighting would let those cells dominate. The same weighting is used for the stratified mean gaps, and the bootstrap resamples items *within* each Level and class so the Level composition is held fixed (`invu4_within.py`). *Second, the sign convention.* We report $|\text{AUC}| = \max(a, 1 - a)$ in the audit tables, so direction is immaterial and 0.5 is chance. The one exception is Table 1, whose inferential comparisons need a signed quantity: a bootstrap difference of folded values would be biased upward wherever a scale sits near chance, which 3B does. That table therefore reports the directional $\text{AUC} = \Pr[c_T > c_F]$, and values below 0.5 there indicate a reversed ordering rather than a weak one.

We use a survival threshold of $|\text{AUC}| \geq 0.62$ (a signal must retain more-than-marginal discrimination after control) and assess family-wise significance with a within-Level max-statistic permutation test.

The threshold itself is a convention, so we report how much of our verdict depends on it (`thresh_sensitivity.py`). The two claims behave very differently. Answer-agreement retains controlled $|\text{AUC}|$ above threshold at every Qwen scale and in the two partial cross-family stress tests for any $\tau \leq 0.642$ (binding run: 14B), so the main claim is one-sidedly robust—lowering the threshold can never break it, and only a threshold above 0.642 would. The confidence claim is not robust in this way: the reported pattern (fails at 3B/14B/Llama/Mistral, passes at 7B/32B) holds only for $\tau \in (0.595, 0.642]$, a window of width 0.047 that happens to contain 0.62. At $\tau = 0.58$ confidence would also “pass” at 14B. We therefore assign gate verdicts to answer-agreement only. Confidence is reported as estimates with intervals, and the threshold-dependent classification it would receive is deferred to Appendix A.5 rather than placed beside the answer-agreement column, where it would invite a comparison the evidence does not support.

Gate 1: within-stratum discrimination. We compute a *within-Level stratified* AUC: pairs of (true, false) items are formed only *within* the same official Level. This removes between-Level difficulty differences, but official Level is a coarse label and within-Level variation remains. If a signal’s raw separability were purely between-level difficulty, its within-stratum AUC would fall to ~ 0.5 . We report a max-statistic, FWER-adjusted permutation p -value by permuting labels within Level (2000 shuffles) and asking how often *any* of the eight signals reaches the focal signal’s observed value.

Gate 2: collider bias. Conditioning on convergence can induce spurious association (a collider). We guard against this by comparing the within-stratum AUC computed on *all* items (using the label “majority answer is wrong,” which is defined for every item, converged or not) to the AUC on the converged subset. If conditioning on convergence *inflates* the signal, the converged AUC is much larger (amplification $\gg 0$); a signal unaffected by this selection shows amplification ≈ 0 or negative.

Gate 3: residual difficulty. We remove the independent estimator D_{ext} from the signal by (i) linear residualization (`−ext`), (ii) additionally removing official Level (`−ext−lvl`), and (iii) fine binning of D_{ext} with within-bin AUC. A signal survives if it stays ≥ 0.62 after the strongest control (`−ext−lvl`).

Gate 4: non-circular control strength. “Survives the control” is only meaningful if the control is strong. We verify that official Level is a strong, non-circular difficulty axis: the false-consensus rate should rise monotonically with Level and Level should itself predict false consensus with high $|\text{AUC}|$. We contrast this with the circular D_{model} (the backbone’s own solo error rate) and with weak structural proxies (question length, operator/number counts, answer magnitude).

4 The Single-Slice Trap

We first reproduce the mistake we want to warn against. On a single hard slice (MATH Level 5 only), round-2 answer-agreement predicts false consensus with $|\text{AUC}| = 0.683$ on a balanced, high-confidence subset ($n = 60$)—a number one might report as a useful detector. But the slice contains only Level-5 items, so the sole within-slice difficulty estimate available is the backbone’s own solo

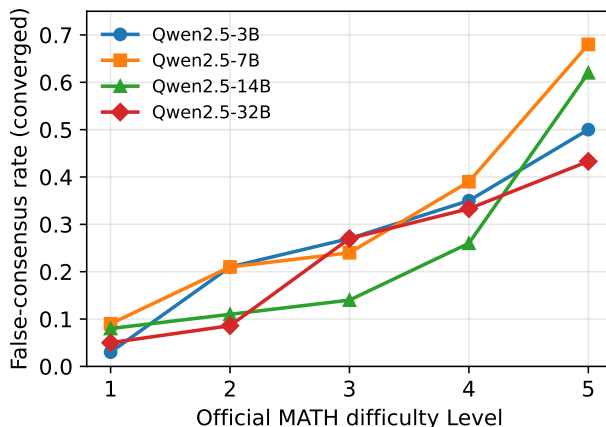


Figure 1: Gate 4 (control strength). False-consensus rate rises monotonically with official MATH difficulty at every scale, confirming official Level as a strong, non-circular difficulty control ($|AUC| = 0.73$ – 0.76 across 3B/7B/14B/32B).

error rate D_{model} , which is circular with the outcome (a model that gets an item wrong alone is both labelled “high difficulty” and likely to be part of a wrong consensus). Linear residualization on that proxy reduces the signal to $|AUC| = 0.514$. This is a diagnostic failure, not a clean estimate of a difficulty-controlled effect: the adjustment is circular, while official Level has zero variance. On the single slice one therefore cannot distinguish “consensus quality” from “difficulty”; the design has removed the axis needed for the test. This is the difficulty-identification trap. The remedy is not more data at Level 5; it is a pool with difficulty *variance* (Section 2) and an explicit audit (Section 3).

5 Results

Gate 4 first: the control is strong and non-circular. Figure 1 shows the false-consensus rate rising monotonically with official Level at every scale, from 3–9% at Level 1 to 43–68% at Level 5. Official Level predicts false consensus with $|AUC| = 0.73$ (3B), 0.75 (7B), 0.76 (14B), 0.73 (32B), all monotone—a strong, non-circular difficulty axis. The circular D_{model} scores higher ($|AUC| = 0.93$ at 3B, 0.91 at 32B) precisely because it is circular; structural proxies (question length etc.) are weak ($|AUC| \leq 0.67$). Official Level is therefore a useful, non-circular control, although its five bands are coarse and cannot remove all within-Level difficulty variation.

Main result: agreement is stable, confidence is measure-dependent. Table 2 collects the audit across scales. It contains answer-agreement only: confidence is not assigned gate verdicts, because its pass/fail labels turn over inside a survival-threshold window of width 0.047 that happens to contain our 0.62 (Section 3). The confidence numbers under the same gates, and the classification they would mechanically receive, are in Table 8 (Appendix A.5); what we claim about confidence is stated below as estimates with intervals and as held-out threshold performance, neither of which needs a survival threshold. *Answer-agreement* clears every specified control in the Qwen–MATH series, at all four scales: its within-stratum AUC is 0.774 (3B), 0.776 (7B), 0.724 (14B), 0.797 (32B)—high, and varying by only 0.073 across an order of magnitude in model size; it stays above the 0.62 survival threshold after the strongest residual control (–ext–lvl: 0.663/0.692/0.655/0.747);

its collider amplification is ≤ 0 ($-0.009/-0.040/-0.038/-0.025$, i.e. it is not inflated by conditioning on convergence); and the within-Level max-statistic permutation test places the observed value outside essentially the whole null distribution ($p < 1/2001$ at 3B, 7B, and 32B, and 0.0010—2 of 2000 shuffles—at 14B, where the folded within-Level value is lowest). We read this as robustness under the stated controls rather than as identification of a difficulty-free quantity; Section 8 states what these controls do and do not rule out.

Confidence behaves differently, and non-monotonically on the measure a deployed monitor would read. Its mean separation $\bar{c}_T - \bar{c}_F$ runs $+0.014$ (3B), $+0.049$ (7B), $+0.024$ (14B), $+0.057$ (32B): a dip at 14B that recovers at 32B. As we show next, that dip is visible only on measures that are sensitive to the reporting scale. At 3B the model is essentially *uncalibrated*: its within-stratum AUC for confidence is 0.512 when reported as $|\text{AUC}|$ —at chance—and it would fail Gate 3 ($-\text{ext-lvl}=0.504$; Table 8). Table 1 reports the same quantity *directionally* as 0.488, i.e. 0.012 on the wrong side of chance; the two differ only by the folding convention $|\text{AUC}| = \max(a, 1 - a)$ used in the audit tables. At 14B the model is *saturated*: both classes sit near the ceiling (0.983 vs 0.959) with a confidence SD of only 0.028, so the within-stratum rank-AUC is retained (0.694) while the absolute separation shrinks to 0.024. At 32B mean false-consensus confidence falls back to 0.919, the spread widens again, and both the separation and the rank AUC reach their highest observed values.

The measure you pick decides the story you tell. A gap of bounded probabilities is mechanically compressed as $\bar{c}_T \rightarrow 1$, and Propositions 1–2 make the consequence precise: a model that merely becomes uniformly more confident *must* show a shrinking probability gap and *cannot* show a changing AUC. We therefore bootstrap (20,000 resamples) every scale under three measures: the probability gap; the unbounded *logit* gap; and the rank AUC, which is invariant to any monotone reparameterisation of the confidence scale (`invu_significance.py`). We add the variance-standardised gap $d = (\bar{c}_T - \bar{c}_F)/\text{SD}(c)$, which separates location from spread (`theory_32b_v2.py`).

Table 1: Four scales, four measures, two estimands (20,000 bootstrap resamples). *pooled* ignores difficulty; *w-Lvl* holds it fixed by Level stratification and resamples within strata (`invu_significance.py`, `invu4_within.py`). The two consecutive arms we test formally are shown for both estimands; that choice was made during analysis, not preregistered, and the remaining contrasts are point estimates. For the 7B→14B dip the three measures agree in *that* narrow sense: it is significant on the bounded probability scale and not detectable on either the rank-invariant measure (AUC) or the variance-normalised one (d), under both estimands. Only AUC is invariant under arbitrary monotone reparameterisation (Proposition 1); d is the weaker check. This agreement is claimed for that arm and those measures only—the 3B→7B rise is precisely where the estimands disagree (n.s. pooled, significant within-Level), so “the measures agree” is not a general property of this table. AUC rows are *directional* ($\Pr[c_T > c_F]$), so values below 0.5 indicate a reversed ordering; the audit tables elsewhere report the folded $|\text{AUC}| = \max(a, 1 - a)$, under which 3B’s 0.488 appears as 0.512. *CI excludes zero.

Measure	estimand	3B	7B	14B	32B	rise (3B→7B)	dip (7B→14B)
probability gap	pooled	+0.014	+0.049	+0.024	+0.057	+0.035 [−0.003, +0.072]	−0.025 [−0.044, −0.006]*
	w-Lvl	−0.013	+0.043	+0.017	+0.047	+0.056 [+0.020, +0.093]*	−0.026 [−0.048, −0.005]*
logit gap	pooled	+0.19	+1.30	+0.93	+1.28	+1.10 [+0.52, +1.75]*	−0.37 [−0.92, +0.17]
	w-Lvl	−0.98	+1.11	+0.63	+1.05	+2.09 [+0.62, +3.03]*	−0.48 [−1.73, +1.02]
standardised d	pooled	+0.16	+0.97	+0.87	+1.26	—	−0.11 [−0.55, +0.35]
AUC (rank)	pooled	0.588	0.774	0.758	0.837	+0.186 [+0.053, +0.321]*	−0.016 [−0.126, +0.093]
	w-Lvl	0.488	0.751	0.694	0.791	+0.263 [+0.123, +0.402]*	−0.057 [−0.188, +0.073]

Three readings follow, stated at the strength the tests support. First, the small-scale end is degenerate, and stratification sharpens rather than softens that: pooled, 3B shows a small positive gap (+0.014), but *within* Level the gap is *negative* (−0.013 on the probability scale, −0.98 on the logit scale with an interval excluding zero) and the rank AUC sits at 0.488, just below chance. The apparently positive pooled gap at 3B is a between-Level artifact—easier Levels carry both more true consensus and higher confidence—so at 3B, holding difficulty fixed, confidence is if anything *anti*-correlated with correctness. We read this as no usable signal rather than as a reliable inversion: only the logit-scale interval excludes zero, and $n_F = 29$ spread over five Levels leaves 453 usable pairs. Second, **the 14B dip appears only on the scale-sensitive measure**, and this is the one conclusion that is stable across both estimands. It is significant on the raw gap (pooled −0.025; within-Level −0.026; both intervals exclude zero) but not detectable on either of the two measures that remove a scale effect (d : −0.11, CI [−0.55, +0.35]; AUC: −0.016 pooled and −0.057 within-Level, both intervals spanning zero). The two remove it in different and unequal senses, and we do not conflate them: AUC is invariant under *any* strictly increasing reparameterisation (Proposition 1), whereas d is computed on the reported confidence scale and is invariant only under a common affine rescaling, so it rules out a pure location-and-spread artifact but not an arbitrary monotone one. A compression account is consistent with this pattern and with the spread—14B’s confidence SD is 0.028, about half of 7B’s 0.051—but four non-significant tests at $n_F = 56$ and 39 cannot establish that discrimination is unchanged, and we do not claim they do. Third, **the 32B point removes the support for an interior peak** rather than disproving one: the separation recovers to +0.057 and AUC reaches 0.837, the highest of the four, with $P(7B \text{ is the global peak}) = 0.105$ on rank and $P(32B \text{ is the maximum}) = 0.834$ (0.240 and 0.719 within-Level, the same direction with less resolution). These are bootstrap resampling frequencies, not posterior probabilities, and the 7B–32B contrast is itself not tested, so this describes where the resampling distribution concentrates rather than a refutation.

We therefore make no claim of an inverted-U, and we also withdraw the weaker “rises then plateaus” phrasing that three points had suggested. What we can say is bounded by the tests: the significant changes in Table 1 are the 3B→7B rise (on the logit and rank scales under both estimands, and on the probability gap under stratification only) and the 7B→14B drop *on the probability scale* (both estimands). The 3B→7B rise on the pooled probability gap is not significant (+0.035, CI [−0.003, +0.072]) though its within-Level counterpart is (+0.056, CI [+0.020, +0.093])—itself a reminder that the estimand, not only the measure, can decide a significance verdict. The standardised gap has no 3B→7B interval because 3B’s spread makes d unstable there. Every other movement, including the recovery at 32B, is a point-estimate trend that our sample cannot resolve. Readers should take the trajectory of confidence discrimination beyond 7B as unresolved, not as established improvement.

Report more than one measure, not merely more scale points. It is tempting to draw the moral that scaling claims need more points, and our own earlier three-point reading did exactly that: with (3B, 7B) the gap rises (“bigger is better calibrated”), with (7B, 14B) it falls (“bigger saturates and degrades”), and we concluded that a third point was needed to see the non-monotonicity. The 32B point shows that this moral was the wrong one. A fourth point does not merely refine the curve—it *reverses* the interior-peak reading, because the 14B dip was never a fact about the model’s discrimination in the first place. Adding points cannot fix that error, since every point would have been measured on the same distorting scale.

The point that does survive is about *measures*, and Propositions 1–2 say why. A bounded probability gap confounds two things a monitor cares about separately: how well the model orders

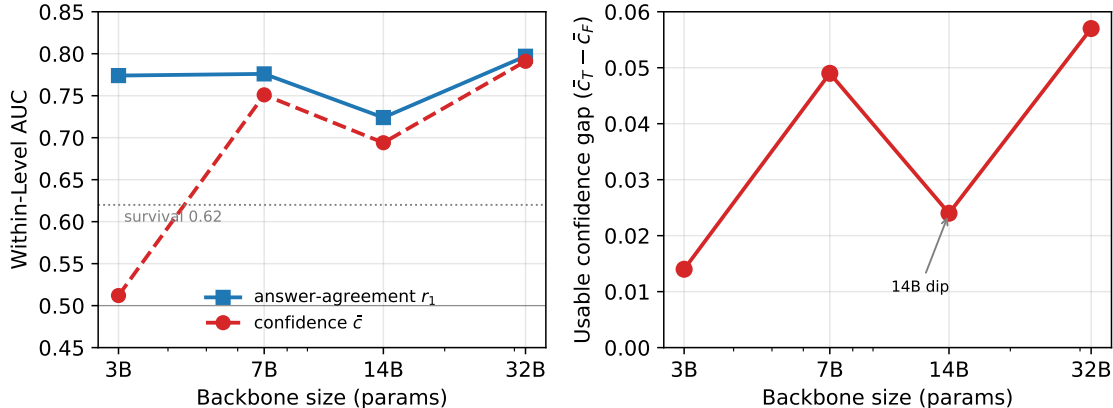


Figure 2: Two signals across four scales. *Left:* within-stratum AUC (difficulty held fixed by stratification), folded as $|AUC|$ to match Table 2; these correspond to the *w-Lvl* rows of Table 1, not the pooled ones. Answer-agreement (blue) stays in a narrow band from 3B to 32B; confidence (red) is at chance at 3B and then tracks it. *Right:* the mean confidence separation $\bar{c}_T - \bar{c}_F$, plotted pooled, dips at 14B and recovers at 32B. Only the 3B→7B and 7B→14B contrasts are tested (Table 1); the remaining movements are point estimates.

items, and how much room the reporting scale leaves to express that ordering. Rank AUC isolates the first; the raw gap mixes both. Because the two can move in opposite directions under nothing more than a location shift, a single-measure scaling study is under-determined no matter how many scale points it has. The recommendation is procedural rather than novel: report a rank-invariant measure alongside the operational one, and treat a divergence between them as a question about the measurement scale rather than an answer about the model. In our own case, reporting only the probability gap would have produced a mid-scale peak; reporting only AUC would have hidden that the separation a threshold actually consumes did shrink at 14B. Neither measure alone would have been enough to notice the ambiguity.

Thresholdability, measured rather than asserted. A mean gap is not a threshold metric, so we do not treat the two as interchangeable. We instead define an operational test (`thresh_perf.py`), whose protocol we fixed before looking at its outcome, though it was added during revision and not preregistered: escalate an item when $\bar{c}$ falls below τ , choose τ on a random half of the items to spend a fixed *false-escalation* budget—the fraction of true-consensus items sent for review—and measure sensitivity (the fraction of false consensus caught) on the held-out half, over 400 stratified splits. Random flagging at the same budget has sensitivity equal to the budget, which gives the baseline. At a 20% budget the median held-out sensitivity is 0.200 (3B), 0.643 (7B), 0.450 (14B), and 0.750 (32B), i.e. lifts over chance of +0.00, +0.44, +0.25, +0.55; the ordering is the same at 10% and 30% budgets. Two things follow. First, at 3B confidence is worthless as a trigger—zero lift—even though its pooled AUC is 0.588, so the small 3B gap is not merely a scale artifact. Second, the 14B dip *does* appear on this genuinely operational metric (+0.44 → +0.25 → +0.55) while it does not appear on AUC. That is the paper’s point in its sharpest form: the observed dip is present for a practitioner choosing this thresholding rule and not detected for one asking whether the model still ranks items correctly. Only reporting both makes the distinction visible. We use “thresholdable” only in this measured sense from here on.

Table 2: Four-gate audit of answer-agreement across model scale. This table reports the audit for the one signal we are willing to assign gate verdicts to; the confidence numbers and their threshold-dependent classification are held in Table 8 (Appendix A.5) and are deliberately not presented as audit outcomes. All quantities are for the false-consensus label on converged items, and all AUC values are *within-Level* (our primary estimand, Section 3), reported as |AUC|. *within* = within-Level stratified AUC (Gate 1, official Level held fixed); *-ext-lvl* = AUC after removing the independent Mathstral difficulty and official Level (Gate 3); *amp* = collider amplification, all-vs-converged (Gate 2; ≤ 0 is good); $p_{\max}$ = the within-Level max-statistic, FWER-adjusted permutation p -value for *this* signal’s own observed value, i.e. the fraction of 2000 label shuffles in which *any* of the eight candidate signals reaches it (1/2001 is the smallest resolvable value). Note this is not the same as the p -value for the per-scale winner: at 14B the highest within-Level value belongs to r_2 -agree (0.789, $p_{\max} < 0.0005$), and evaluating the max statistic at r_1 -agree’s lower value of 0.724 costs resolution (`_fwer_r1.py`).

Backbone	within	-ext-lvl	amp	$p_{\max}$	verdict
Qwen2.5-3B ($n_c=124$)	0.774	0.663	-0.009	< 0.0005	pass
Qwen2.5-7B ($n_c=189$)	0.776	0.692	-0.040	< 0.0005	pass
Qwen2.5-14B ($n_c=175$)	0.724	0.655	-0.038	0.0010	pass
Qwen2.5-32B ($n_c=178$)	0.797	0.747	-0.025	< 0.0005	pass

Cross-family stress test. To test whether the answer-agreement result is a Qwen artifact, we re-run the same protocol on two independent families, Llama-3.1-8B and Mistral-7B (Table 3). Answer-agreement remains high in both under a weaker, only partially applicable control analysis; this is supporting evidence, not a second run of the full Qwen audit. Answer-agreement retains its within-Level discrimination on each—AUC of 0.70 (Llama) and 0.73 (Mistral), in the same band as Qwen’s 0.72–0.80—and on both it stays above threshold after the strongest control we ran there (*-str-lvl* of 0.695 and 0.691), with within-Level permutation p of 0.005 and 0.018. Two limits keep this short of a replication of the Qwen audit. First, the cross-family runs use the structural rather than the Mathstral control, so they exercise a weaker version of Gate 3 than the Qwen scale series. Second, on Mistral the difficulty axis itself is *not monotone* (false-consensus rate 0.40/0.53/0.62/0.57/1.00 across Levels 1–5, with only 7–8 items in the top two cells), so Gate 4’s premise—that official Level is a well-behaved difficulty ruler—is only partly met there. We therefore read these two families as evidence that high agreement discrimination is not unique to Qwen, not as additional four-gate passes. For context, confidence is near chance on both non-Qwen families (within-Level 0.56 / 0.52, and 0.545 / 0.508 after removing structural proxies and Level). We report those estimates descriptively and do not assign them gate verdicts. A further cross-family observation: the three families converge at very different rates (Mistral only $\sim 29\%$ of items reach majority agreement, vs. Llama’s higher false-consensus-heavy regime, $r_{fc} = 0.67$), yet the controlled agreement estimates remain high—suggestive that the result tracks the signal rather than a particular model’s agreement dynamics, though three families cannot settle that.

Heterogeneous-agent stress test. We also test three mixed systems, each combining a Qwen backbone with Llama-3.1-8B and Mistral-7B. To conserve compute while preserving an auditable counterfactual, their round-1 samples are fixed matched draws from the completed homogeneous runs; only the post-exchange round-2 revisions are newly generated. The statistic below is round-1 agreement, so this is a constructed heterogeneous round-1 stress test, *not* an estimate of the effect of answer exchange or a fully fresh end-to-end heterogeneous replication. Two analyzable cells retain substantial within-Level agreement discrimination (Qwen7+Llama8+Mistral7: 0.779 [0.671, 0.869]; Qwen14+Llama8+Mistral7: 0.727 [0.630, 0.814]). The Qwen3 mixture has only 13 false-consensus

Table 3: Cross-family stress test on MATH. Round-1 answer-agreement under the same protocol in three model families. *within* = within-Level stratified |AUC|; *−str−lvl* = |AUC| after removing structural proxies and official Level; $p_{\max}$ = max-statistic, FWER-adjusted permutation p -value. The two non-Qwen runs use a weaker control set than the Qwen scale series: structural proxies replace independent Mathstral difficulty, and Mistral’s Level axis is non-monotone with only 7–8 items in its top two cells. We therefore report estimates rather than four-gate verdicts. (n_c : converged items; r_{fc} : false-consensus rate among converged.)

Family	n_c	r_{fc}	within	−str−lvl	$p_{\max}$
Qwen2.5-7B	189	0.30	0.776	0.692	< 0.0005
Llama-3.1-8B	102	0.67	0.703	0.695	0.005
Mistral-7B	68	0.56	0.729	0.691	0.018

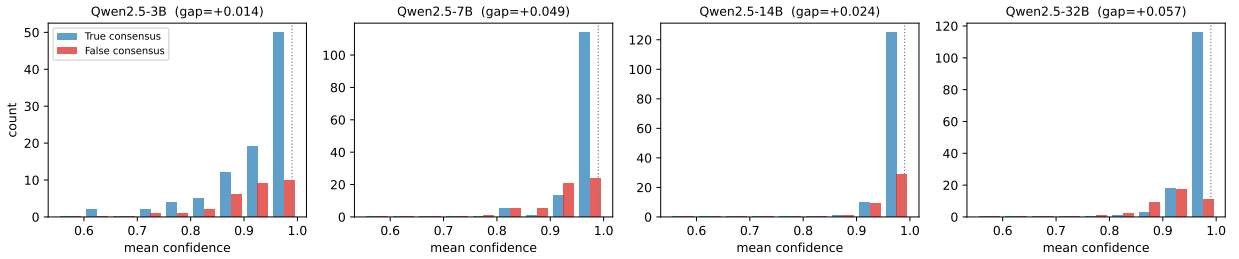


Figure 3: Confidence distributions by scale. Confidence distributions of true (blue) vs. false (red) consensus on converged items. At 3B and 14B the false set lies entirely inside the true range (overlap = 1.00)—uncalibrated at 3B, ceiling-saturated at 14B (dotted line at 0.99). The 7B and 32B runs show larger absolute gaps. Range containment does not by itself mean a threshold is useless: at a 20% false-escalation budget the held-out lift is +0.00 at 3B but +0.25 at 14B, so only the 3B collapse defeats thresholding outright.

cases and is explicitly underpowered (0.615 [0.490, 0.734]). These data weaken the concern that the round-1 MATH pattern is limited to homogeneous same-family samples, but they do not establish a heterogeneity or exchange effect, and do not reproduce all four gates. As throughout, these intervals are conditional on this one archived generation trace.

Why confidence fails: calibration vs. saturation. Figure 3 makes the two failure mechanisms concrete by plotting the confidence distributions of true and false consensus at each scale. At 3B the two overlap almost completely: every false-consensus item’s confidence lies inside the true-consensus range (overlap = 1.00), and the mean gap is only +0.014—the model is simply uncalibrated. At 14B both distributions are pushed against the ceiling ($\bar{c}_T = 0.983$, $\bar{c}_F = 0.959$; 58% of true and 13% of false items report ≥ 0.99), so again the false set is entirely contained in the true range (overlap = 1.00), leaving a fixed threshold little absolute margin to work with—though, as the held-out test above shows, not none (+0.25 lift at a 20% budget, against +0.00 at 3B). At 7B the overlap dips (0.96) and the separation widens to +0.049. The two patterns look alike on the mean but differ in the distributions: 3B carries no usable ordering, whereas 14B retains one inside a narrow band. Consistent with that reading, the same family at 32B de-saturates ($\bar{c}_F$ back to 0.919) and shows the widest separation of the four points. We emphasise that this is a descriptive contrast between distributions, not a tested claim about which mechanism operates at each scale.

Table 4: Cross-domain point estimates for answer-agreement. Within-stratum $|AUC|$ of round-1 answer-agreement (r_1 -agree) for false-consensus prediction, at ~ 7 –8B scale. ARC uses the official Easy/Challenge split as the difficulty axis; MATH uses Levels 1–5. All six point estimates exceed the survival threshold, but this is not six independent audit passes. The evidence is not equally strong across columns: the within-Level permutation tests reported in Section 5 cover the MATH column, whereas for ARC we have only bootstrap intervals (`_arc4pt.py`), shown here in brackets. Every ARC interval has a lower end above 0.5, but three of them extend below the 0.62 survival threshold, so that column should be read as suggestive rather than as a second audit.

Family	MATH (numeric)	ARC (science MCQ)
Qwen2.5-7B	0.776	0.713 [0.599, 0.834]
Llama-3.1-8B	0.703	0.667 [0.567, 0.777]
Mistral-7B	0.729	0.797 [0.715, 0.882]

Table 5: New P0 stress tests. Heterogeneous MATH systems use fixed matched homogeneous round-1 draws; round-2 revisions after cross-family answer exchange are newly generated but are *not* the outcome in this table. MMLU is a disjoint 200-item holdout with a shared short MCQ prompt; its STEM/humanities grouping is only a coarse stratum. Values are within-stratum $|AUC|$ for round-1 agreement with item-bootstrap intervals conditional on one archived generation trace. “Underpowered” denotes fewer than 15 false-consensus items.

Setting	items	converged / false	within-stratum $ AUC $
Qwen3+Llama8+Mistral7 (MATH)	196	76 / 13	0.615 [0.490, 0.734]
Qwen7+Llama8+Mistral7 (MATH)	199	93 / 20	0.779 [0.671, 0.869]
Qwen14+Llama8+Mistral7 (MATH)	197	83 / 20	0.727 [0.630, 0.814]
Qwen2.5-3B (MMLU)	200	199 / 70	0.568 [0.511, 0.629]
Qwen2.5-7B (MMLU)	200	200 / 55	0.535 [0.487, 0.586]
Qwen2.5-14B (MMLU)	200	200 / 45	0.566 [0.521, 0.622]
Llama-3.1-8B (MMLU)	200	186 / 62	0.637 [0.572, 0.702]
Mistral-7B (MMLU)	200	197 / 86	0.612 [0.562, 0.666]

Cross-domain generalization, and where it stops. Finally, we test whether the MATH point estimates transfer beyond numeric math. On ARC (4-way science multiple-choice, official Easy/Challenge split as the difficulty axis), answer-agreement has within-stratum point estimates of 0.67–0.80 across the three families, in the same range as MATH (Table 4). This is suggestive rather than a clean replication: the ARC column uses bootstrap intervals rather than max-statistic permutation tests, and all three intervals extend below the 0.62 survival threshold. One instructive contrast is descriptive: on ARC, Mistral confidence has $AUC = 0.70$, whereas its MATH estimate is 0.52. The present analysis does not establish whether task format or another domain difference causes that gap.

The disjoint MMLU holdout makes the domain boundary less dependent on a single model. The original 32B MMLU stress test gave a weak within-tier estimate of 0.581 [0.506, 0.675]. On the new disjoint 200-item holdout, all five general-purpose models at or below 14B are much weaker than their MATH runs: 0.535–0.637 (Table 5). The three Qwen cells are close to chance (0.535–0.568); Llama and Mistral are higher but remain far below their MATH estimates. Because the holdout uses a short MCQ protocol and a coarse two-tier proxy, neither cross-domain gaps nor apparent family differences are causal estimates. They are nevertheless a robustness check of the qualitative boundary across five models, with 45–86 false cases per cell rather than a single 24-case observation.

Table 6: Independent fresh-generation repetition of the P0 configurations. The reported values are within-stratum $|AUC|$. Each difference interval is a paired item bootstrap conditional on the original and repeated archived traces; it is not a confidence interval over the model’s stochastic-generation distribution. Usable MATH counts differ across traces because length-truncated outputs are excluded.

Setting	original $ AUC $	repeat $ AUC $	repeat – original, 95% CI
MATH / Qwen2.5-3B	0.774	0.701	−0.073 [−0.155, +0.089]
MATH / Qwen2.5-7B	0.776	0.715	−0.061 [−0.139, +0.042]
MATH / Qwen2.5-14B	0.724	0.769	+0.046 [−0.016, +0.181]
MMLU / Qwen2.5-3B	0.568	0.521	−0.048 [−0.111, +0.012]
MMLU / Qwen2.5-7B	0.535	0.528	−0.006 [−0.067, +0.051]
MMLU / Qwen2.5-14B	0.566	0.538	−0.028 [−0.083, +0.024]
MMLU / Llama-3.1-8B	0.637	0.649	+0.013 [−0.076, +0.104]
MMLU / Mistral-7B	0.612	0.573	−0.039 [−0.096, +0.016]

Fresh-generation repetition. To distinguish a fragile archived trace from a pattern that reappears once, we generated a fully fresh trace for every P0 configuration: Qwen2.5-3B/7B/14B on MATH and all five models on the disjoint MMLU holdout. The MATH repeat estimates are 0.701, 0.715, and 0.769, against 0.774, 0.776, and 0.724 in the original traces (Table 6). The MMLU repeat estimates are 0.521/0.528/0.538 for Qwen-3B/7B/14B, 0.649 for Llama-8B, and 0.573 for Mistral-7B; thus the near-chance Qwen MMLU pattern and the modestly higher non-Qwen values recur. All paired item-bootstrap differences span zero. This is evidence of agreement between two archived stochastic traces, not a claim that the configurations are statistically identical or that we have estimated run-to-run generation variance. In particular, the MATH repeats contain 225/235/238 usable items because length-truncated outputs are excluded, so the comparison is deliberately a trace-robustness check rather than a fixed-item causal contrast.

The agreement patterns suggest a mechanism, and it is the one Proposition 3 points to. On MATH, false-consensus items are unanimous only 20% of the time against 78% for true consensus—disagreement exposes the error. On MMLU the two rates are 75% and 92%: when the model is wrong, all three agents still usually give the *same* wrong answer. A natural reading is that agreement detects *idiosyncratic reasoning slips*, which three independent samples resolve differently, but not *shared knowledge gaps*, where the error is systematic and reproduces across samples. We offer this as a hypothesis consistent with three domains, not as an established taxonomy of MATH, ARC, and MMLU errors; one knowledge-recall domain is not a survey. Section 6.2 makes the quantity behind it explicit.

We flag one further cell as untestable rather than negative: ARC at 32B produces only 7 false-consensus items in 240 (0 on ARC-Easy), so its point estimate ($AUC = 0.635$, CI [0.504, 0.833]) spans half the range and we do not interpret it. That is a fact about the benchmark being too easy for the model, not evidence either way.

Robustness details. Gate 3 uses an independent Mathstral-7B difficulty estimator with $|AUC|$ of 0.77–0.78 against Qwen error—a fine-grained, non-circular control finer than the five Levels. After removing it (and Level), answer-agreement remains above threshold at all four scales. The corresponding confidence estimates are above threshold at 7B and 32B and below it at 3B and 14B, but we do not treat those threshold-sensitive classifications as audit verdicts. Gate 2 flags agree-slope as a collider artifact at 7B (amplification +0.103), confirming the test has discriminating power and is not a rubber stamp; answer-agreement’s amplification is non-positive at every Qwen scale. For r_1 -agree, the max-statistic adjusted p -value—the fraction of 2000 within-Level

permutations in which any of the eight signals reached the observed value was 0 at 3B, 7B and 32B (i.e. $p < 1/2001$, the resolution limit of 2000 shuffles) and 2/2000 at 14B; we report these bounds rather than 0.000, which 2000 shuffles cannot resolve. This permutation family covers the eight candidate signals only, and does not correct for our selection of scales, domains, and controls.

6 What the Signals Are, Mathematically

Two of our empirical claims have exact mathematical content, and separating that content from the rest is what keeps the claims honest. The first is a pair of one-line invariance facts that determine *which* of our confidence measurements can possibly be evidence of a capability change (Section 6.1). The second is a sampling identity showing that answer-agreement is informative through a small number of estimable quantities, which converts our domain-boundary finding from a narrative into a measurement (Section 6.2). We state nothing here that we cannot either prove in two lines or expose to a test it can fail.

6.1 Invariance: which confidence measurements can carry evidence

Let a converged item carry a latent evidence score z , and let the reported confidence be $c = s(z)$ for some strictly increasing *verbalisation map* s . Write T and F for true and false consensus.

Proposition 1 (Rank functionals are invariant). *For any strictly increasing g , $\text{AUC}(g(c)) = \text{AUC}(c)$ exactly.*

Proof. AUC depends on c only through the events $\{c_T > c_F\}$ and $\{c_T = c_F\}$, and a strictly increasing g preserves both. $\square$

Proposition 2 (Mean gaps are not invariant). *The gap $D_p = \mathbb{E}[c \mid T] - \mathbb{E}[c \mid F]$ depends on s . Under a pure location shift $z \mapsto z + t$, AUC is constant in t , while if s is bounded with $s' \rightarrow 0$ at the ceiling then $D_p(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Proof. Invariance of AUC is Proposition 1 applied to the shift. For D_p , both conditional means are squeezed into the shrinking interval $[s(t + \min z), \sup s]$, whose width tends to 0. $\square$

The consequence is a logical one, and it is the reason we report three measures rather than one. “*The probability gap shrank*” and “*the model got worse at ranking*” are independent statements. A model that merely becomes uniformly more confident—a location shift, with no change whatsoever in how it orders items—*must* exhibit a shrinking probability gap and *cannot* exhibit a changing AUC. A shrinking gap is therefore evidence of lost discrimination only if it also appears on a *rank-invariant* measure. (We reserve that term for AUC: it is the only one of our measures invariant under any strictly increasing reparameterisation. The variance-standardised gap of Section 5 removes a common affine rescaling and no more, so it is a weaker check, not a second rank-invariant one.) This is what motivates the standardised decomposition in Section 5, and it is why we decline to describe the 14B behaviour as degraded capability: the drop is significant on the raw gap and not detectable on either AUC or the variance-standardised gap (Table 1). We verify Proposition 1 numerically as an implementation check—applying identity, logit, x^5 , and e^{3x} to every confidence value leaves each scale’s AUC unchanged to six decimals (`theory_32b.py`).

6.2 Agreement under a restricted sampling model

For a fixed item, model the $k = 3$ agents as i.i.d. draws from that item’s answer distribution q over K candidate answers, and let $m = \max_k q_k$ be the *top mass*. Assigning the remaining $1 - m$ uniformly over the other $K - 1$ options ($r = (1 - m)/(K - 1)$), the three observable agreement patterns have probabilities

$$\begin{aligned} \Pr[\text{unanimous}] &= m^3 + (K - 1)r^3, & \Pr[2-1] &= 3[m^2(1 - m) + (K - 1)r^2(1 - r)], \\ \Pr[\text{all distinct}] &= 1 - \Pr[\text{unanimous}] - \Pr[2-1], \end{aligned} \quad (1)$$

on which the feature $r_1\text{-agree}$ takes the values 1, 2/3, 1/3.

Proposition 3 (Under the uniform-tail model, agreement is informative only through top mass). *Assume the answer distribution has the restricted form $q = (m, r, \dots, r)$ with $r = (1 - m)/(K - 1)$, i.e. one preferred answer and a uniform tail over the remaining $K - 1$ options. Then the agreement distribution (1) depends on q only through m . Hence if $m_F = m_T$, the distribution of $r_1\text{-agree}$ is identical on true and false consensus and $\text{AUC} = 1/2$ exactly.*

Proof. Immediate from (1): under the uniform-tail form all three probabilities are functions of m (and the fixed K) alone. $\square$

The uniform-tail restriction is not a technicality—it is the entire reason the reduction to one scalar works. For an unrestricted q the claim is false: two distributions can share a top mass and still differ in how the remaining mass is concentrated, giving different unanimity, 2–1, and all-distinct probabilities. We state the proposition in restricted form for that reason, and we test the restriction rather than assume it (Table 7 and the failure mode below). Note also that the predicted AUC depends on m_T , m_F , and K separately, not on the difference $m_T - m_F$ alone; equal top masses force chance, but a given difference need not give the same AUC at different absolute levels.

Proposition 3 is elementary, but it is the useful kind of elementary: within this restricted sampling model it replaces the vague question “does answer-agreement generalise across domains?” with quantities we can estimate—the two top masses m_T and m_F —and specifies the exact condition under which the signal must fail. It also supplies the mechanism. When errors are *idiosyncratic reasoning slips*, three samples of a wrong item diverge, m_F is small, and disagreement exposes the error. When errors are *shared knowledge gaps*, the model does not know the fact and all three samples repeat the same wrong answer, m_F is large, and agreement cannot distinguish shared correctness from shared ignorance.

A test the model can fail. Equation (1) would be vacuous if we fit it to the quantity we then report. We therefore fit *one* number per group— m_g , recovered by inverting the unanimity rate—and use it to *predict* two things that never entered the fit: the 2–1 versus all-distinct split, and the AUC of agreement. The three-category pattern distribution has two degrees of freedom per group, while the fitted model spends only one, so the remaining degree of freedom plus the AUC are genuine out-of-fit predictions. We note the check is weaker than it looks: $r_1\text{-agree}$ takes only three values and its AUC is dominated by the same unanimity contrast used for fitting, so agreement between predicted and observed AUC is not fully independent evidence.

Table 7 reports the outcome. The predicted AUC matches the observed one to within 0.006 in all three domains, and a label-shuffle control drives the prediction to 0.498–0.503, which is a pipeline sanity check but not an independent power analysis. The fitted top masses show an ordering consistent with the restricted model: MATH and ARC have a large true–false separation

Table 7: Top mass predicts whether agreement works. For each domain we fit *one* number per group—the top mass m_g , recovered by inverting the observed unanimity rate—and then predict the AUC of answer-agreement, which never entered the fit. Prediction error is within 0.006 everywhere, while a label-shuffle control collapses the prediction to chance, providing a pipeline sanity check. The domain ordering is consistent with the restricted model: a large true-false separation in fitted top mass accompanies a strong signal, whereas a small separation accompanies a near-chance signal. The proposition itself guarantees only the equal-top-mass case; it does not establish this ordering in general. The AUC columns here are *pooled* over items, not within-Level: the sampling model of Eq. (1) is a statement about the answer distribution of the whole set, not a difficulty-controlled contrast, so the pooled value is the one the model predicts. They are therefore not comparable to the within-Level values in Table 2 or Table 4 (e.g. MATH 32B is 0.827 pooled and 0.797 within-Level). MMLU’s fitted $m_F = 0.909$ is a model-consistent diagnostic: under the uniform-tail approximation, wrong answers are highly concentrated and the three samples repeat the same error. Reproduce with `theory_32b_v2.py`.

Domain (32B unless noted)	fitted top mass			agreement AUC			
	m_F	m_T	$m_T - m_F$	observed	predicted	error (95% CI)	shuffled
MATH	0.585	0.919	+0.334	0.827	0.820	−0.006 [−0.027, +0.018]	0.503
ARC (Mistral-7B)	0.669	0.969	+0.301	0.807	0.810	+0.003 [−0.004, +0.010]	0.498
MMLU	0.909	0.973	+0.064	0.584	0.585	+0.001 [+0.000, +0.005]	0.501

(+0.334, +0.301) and a strong pooled signal, whereas MMLU has a small separation (+0.064) and a signal near chance. Proposition 3 guarantees chance only when the two top masses are equal; it does not prove a monotone relation in their difference. The diagnostic number is the model-implied $m_F = 0.909$ on MMLU: to reproduce the observed unanimity rate on wrong items, the fitted distribution must place 91% of its mass on the preferred answer, so three agents repeat the same error. For MATH the corresponding figure is $m_F = 0.585$. These are quantities implied by the restricted model, not measured per-item top masses: because unanimity is nonlinear in item-level top mass, inverting a pooled unanimity rate over heterogeneous items does not recover their average. Because MATH has no natural option count, we verified that the recovered m_F is insensitive to K (0.576–0.585 for $K \in \{4, 10, 30, 100\}$); this is numerical stability of the inversion, not evidence that m_F is a real per-item probability.

What this model does not capture. The AUC prediction succeeds, but the full pattern distribution does not: on MATH the predicted 2–1 rate for false consensus is 0.444 against an observed 0.350. The uniform-tail assumption in (1) is clearly imperfect, and the AUC prediction survives partly because AUC is driven mainly by the unanimity contrast. We therefore claim only that top mass predicts the *discriminative power* of agreement, not that (1) is a correct model of the answer distribution.

A proposition we withdraw. While developing this section we conjectured a bound relating the two scales of Section 6.1, $|D_p| \leq (b - a)(2\text{AUC} - 1)$ for confidence supported on $[a, b]$, which would have quantified how much a bounded scale can distort a gap at fixed ordering. A randomised search over discrete distribution pairs falsified it (9911 violations in 60,000 trials). We report this because the search was part of our protocol for this section—every claimed inequality was subjected to counterexample search before being written down—and because the absence of such a bound is itself informative: the raw gap and the rank measure can diverge without limit, which is precisely why both must be reported.

7 Related Work

Multi-agent failure modes. Recent studies document that multi-agent debate and role-based collaboration often fail to beat strong single-agent baselines and can converge prematurely on wrong answers [Cemri et al., 2025, Wu et al., 2025]. Other work studies how confidence, uncertainty, and agent diversity affect debate dynamics or can improve communication [Yoffe et al., 2025, Zhu et al., 2026]. MAST [Cemri et al., 2025] provides a taxonomy of such failures but diagnoses them offline with human annotation; we target the online, ground-truth-free detection question and, crucially, ask whether the natural signals for it are real or difficulty artifacts.

Efficiency-oriented agent optimization. DyLAN, AgentPrune, and GPTSwarm [Liu et al., 2024, Zhang et al., 2024, Zhuge et al., 2024] optimize which agents communicate or the interaction topology for cost, not consensus correctness. Our object of study is orthogonal: a monitoring signal for consensus quality, and its validity under difficulty confounding.

Ground-truth-free uncertainty quantification. Self-consistency [Wang et al., 2023] aggregates multiple chain-of-thought samples by majority vote and uses voting entropy as an uncertainty proxy; semantic uncertainty [Kuhn et al., 2023] clusters semantically equivalent outputs to refine entropy estimates. Both target single-model uncertainty over multiple generations. We study the multi-agent consensus scenario and, critically, audit whether these signals (including agreement and confidence) are genuine reliability indicators or just difficulty proxies after controlling for problem difficulty with non-circular estimators.

Confidence and calibration. Self-reported and elicited confidence is known to be imperfectly calibrated and to change with scale and tuning [Kadavath et al., 2022, Tian et al., 2023]. Our contribution is not a new calibration method. For the specific task of separating true from false consensus after our difficulty controls, the raw confidence gap and held-out threshold performance dip at 14B and recover at 32B, whereas the rank-based measure shows no detectable 14B dip. This measure dependence is a caution against inferring a scaling law from a bounded confidence scale.

Benchmarks. Our mathematical-reasoning experiments use MATH [Hendrycks et al., 2021b]; the cross-domain stress tests use ARC [Clark et al., 2018] and MMLU [Hendrycks et al., 2021a]. These datasets provide different notions of difficulty and answer structure, so we treat cross-domain comparisons as stress tests rather than interchangeable replications.

Confounding and evaluation validity. Difficulty and selection confounds are recurring threats in ML evaluation [Geirhos et al., 2020]. Our methodological point—that a single-difficulty slice makes the only within-slice difficulty estimate circular, so a difficulty-independent claim is untestable there—is specific to consensus-quality signals but echoes general concerns about collider bias from conditioning and about controls that are too weak to license “survives control” conclusions.

8 Discussion and Limitations

Practical guidance. For a practitioner building an online consensus monitor, the results give a candidate to audit first and a caveat. The candidate: *answer-agreement*. On Qwen–MATH its discrimination held up under every specified control, stayed within a narrow band across an order of magnitude of scale, and requires no confidence calibration. The caveat: *check that your errors*

are the right kind first. Agreement works by letting independent samples disagree, so it is most informative when errors are idiosyncratic and least informative when they are shared. The check needs a small labelled audit set rather than none—compare the unanimity rate on items the system gets wrong against items it gets right. If the two are close (as on MMLU, 75% vs 92%), agreement is unlikely to be useful in that domain regardless of model size. *Confidence* is worth adding when the backbone is neither uncalibrated nor saturated; near the ceiling the classes are separated by so little absolute margin that a fixed threshold has little room to work, even where rank discrimination is retained.

Report rank-invariant and operational measures together. Our methodological recommendation is not that scaling studies need more points. It is that they need more than one *measure*. The underlying invariance facts are standard measurement theory, not a new result; what our data add is a concrete case where ignoring them would have produced a confident and wrong scaling story. A bounded operational measure such as a probability gap mixes two things: how well the system orders cases, and how much room the reporting scale leaves to express that ordering. Propositions 1 and 2 note that these can move in opposite directions under nothing stronger than a location shift, so a single-measure study is under-determined at any number of scale points—adding points measured on the same distorting scale cannot repair it. We recommend reporting a rank-based measure alongside the operational one and treating a divergence between them as a question about the measurement scale rather than an answer about the system. We note one limit of this advice: rank AUC is free of the reporting scale, but it is not free of item composition, ties, or the selection induced by conditioning on convergence.

Limitations. (1) *Scale curve is single-family; cross-family is single-scale.* The four-point curve is established within the Qwen2.5 family on MATH. We obtain positive answer-agreement point estimates under weaker controls on two other families (Llama-3.1-8B, Mistral-7B), but each at a single ~ 7 –8B scale. Those runs are stress tests, not full four-gate replications, and cannot trace the confidence curve within either family. (2) *Non-significant is not absent.* Our claim that the 14B dip does not appear on the rank-invariant or variance-normalised measures rests on two non-significant tests with $n_F \in [39, 56]$ (Table 1). That is evidence of no *detectable* drop, not proof of no drop; a substantially larger collection could resolve a small effect we cannot see. We are careful to phrase the conclusion as an argument against *single-measure* inference rather than as a positive claim that discrimination is exactly flat. (3) *The domain boundary has broader but still non-identifying support.* The original 32B MMLU result is now complemented by a disjoint 200-item holdout run on five general-purpose models at or below 14B (within-tier $|\text{AUC}| = 0.535\text{--}0.637$, 45–86 false-consensus cases per cell), with a fresh-generation repeat of all five cells (0.521–0.649). Its short MCQ prompt differs from the MATH protocol and its STEM/humanities grouping is a coarser difficulty proxy than MATH’s five Levels or ARC’s official split. Thus the replicated weakness is a two-trace domain-stress observation, not a direct causal domain or model-family comparison or an estimate of generation-level variance. The mechanism (Proposition 3, $m_F = 0.909$) is consistent across the three domains we measured, but “knowledge-recall domains break agreement” is a hypothesis supported by one such domain, not a survey. (4) *One cell is untestable, not negative.* ARC at 32B yields only 7 false-consensus items in 240, so we exclude it from interpretation entirely rather than report its uninformative point estimate. (5) *Residual-difficulty ceiling.* The independent Mathstral estimator approaches its own ceiling at Levels 4–5, so the finest difficulty control is weaker in the hardest band; we mitigate with official-Level stratification but note the residual gate is strongest at low-to-mid difficulty. (6) *Homogeneous stochastic replicas.* The three “agents” are

stochastic samples from the same checkpoint with the same prompt, not independently trained or role-specialised agents. Temperature is 0.7 and the generation cap is 1536 tokens. The fixed seed controls item ordering, not backend sampling, so exact generations may vary. We add an answer-exchange stress test with mixed Qwen/Llama/Mistral systems, but its round-1 draws are reused from homogeneous runs; fully fresh heterogeneous-agent, prompt/decoding, and frontier-model robustness remain untested. (7) *Moderate false-consensus counts*. Converged false-consensus counts are 29/56/39/40 (3B/7B/14B/32B). Max-statistic, FWER-adjusted permutation p -values for r_1 -agree are at or near the resolution floor of 2000 shuffles ($p_{\max} < 1/2001$ at three scales, 0.0010 at 14B), but the smallest cells contribute little stratified signal; we report within-Level pair counts in the audit outputs and avoid over-interpreting per-Level point estimates.

9 Ethics Statement

The experiments use public benchmark questions and model outputs; they involve no human participants or personal data. The principal risk is operational over-trust: agreement can reflect shared error rather than correctness, so the signal should not be deployed as a substitute for labelled validation or human review in consequential settings. We report model sizes, decoding settings, sample counts, and exclusions to support scrutiny of compute use and selection effects.

10 Conclusion

A cheap, ground-truth-free signal for multi-agent false consensus looks attainable, but only if one first avoids the difficulty-confounding trap that makes a naive single-slice round-2 agreement detector look strong ($|AUC|=0.683$) while a circular own-model error adjustment reduces it to 0.514. This is evidence that the single-slice design cannot identify the difficulty-independent claim, not a valid deconfounded estimate. Our four-gate audit tests whether a signal’s predictive power survives explicit difficulty controls, and applied across four model scales it finds answer-agreement stable under all of them (within-stratum $AUC=0.72$ – 0.80 across an order of magnitude). We stress that this is robustness under the controls we ran, not identification of a difficulty-free quantity.

Two boundaries make that recommendation usable rather than merely encouraging. First, a boundary on *measurement*: the apparent mid-scale peak in confidence usability, which three points had suggested and which our fourth point removes, was visible only on a bounded scale. Since rank functionals are invariant under monotone reparameterisation and mean gaps are not, the two can diverge without any change in the underlying ordering—so reporting one measure is under-determined at any number of scale points. Second, a candidate boundary on *domain*: agreement works by letting independent samples disagree, and it is substantially weaker on two MMLU pools, across the original 32B stress test and five new ≤ 14 B systems, where wrong answers are reproduced rather than varied. A one-parameter sampling model ties this to the concentration of the answer distribution and suggests a labelled-audit diagnostic. We release the audit protocol, the pools, the independent difficulty estimator, and all runs to make these claims checkable.

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A Details and Negative Results for Section 6

A.1 Proof details

Proposition 1 (rank invariance). Let g be strictly increasing. For any pair (c_T, c_F) we have $g(c_T) > g(c_F) \iff c_T > c_F$ and $g(c_T) = g(c_F) \iff c_T = c_F$. Since $\text{AUC} = \Pr[c_T > c_F] + \frac{1}{2} \Pr[c_T = c_F]$ depends on the joint law only through these two events, it is unchanged. Note this is exact, not asymptotic, and holds for the empirical AUC as well: it is a statement about the ordering of a finite sample. This is why we can use it as an implementation check.

Proposition 2 (mean gaps are not invariant). Write $c = s(z)$ with s increasing, bounded above by $M = \sup s$, and $s'(x) \rightarrow 0$ as $x \rightarrow \infty$. Under $z \mapsto z + t$, $D_p(t) = \mathbb{E}[s(z_T + t)] - \mathbb{E}[s(z_F + t)]$. Both terms lie in $[s(t + z_{\min}), M]$, and $s(t + z_{\min}) \rightarrow M$ because s is increasing and bounded, so the interval width tends to 0 and hence $D_p(t) \rightarrow 0$. The AUC is unchanged for every t by Proposition 1 applied to $g(x) = x + t$ composed with s . Thus the same ordering is compatible with any gap in $(0, D_p(0)]$, which is the formal content of “the gap mixes ordering with scale”.

Numerical check of Proposition 1. Applying identity, logit, x^5 , and e^{3x} to every confidence value leaves the per-scale AUC identical to six decimals: 0.588203 (3B), 0.774235 (7B), 0.758201 (14B), 0.836685 (32B) under all four maps (`theory_32b.py`). Any deviation would have indicated a tie-handling bug in our AUC implementation; there is none.

A.2 A rescaling test that does not work, and why

Our first attempt to show that the 14B dip is a scale effect was to map every scale’s confidences through one common monotone ruler—the pooled empirical CDF—and check whether the dip shrank. It did not; it grew, from -0.025 to -0.066 . We report this because the failure is instructive rather than data-driven: *the test does not test what it claims*. A pooled CDF is a monotone map, so by Proposition 1 it leaves every scale’s AUC exactly unchanged (we verified this numerically), while the transformed *means* still depend on the chosen ruler. Rescaling therefore replaces one arbitrary D_p with another and cannot remove compression. The correct decomposition is to divide by a spread estimate, which is what Section 5 reports as d .

A.3 A conjectured inequality, falsified

We conjectured that a bounded scale can distort a gap only in proportion to the ordering it expresses: for confidence supported on $[a, b]$,

$$|D_p| \leq (b - a)(2\text{AUC} - 1). \quad (2)$$

Had it held, it would have given a quantitative ceiling on how much of a gap change can be attributed to compression at fixed ordering. It is false. A randomised search over pairs of discrete distributions on a common support (2–5 atoms, uniform random masses and locations, 60,000 trials) found 9911 violations, with a worst-case excess of $+0.297$ (`theory_32b.py`). Intuitively, AUC can be near $1/2$ while the two means differ substantially, because AUC ignores *how far apart* the atoms are. We

drop (2) rather than patch it. Its absence is itself informative: without such a bound, the raw gap and the rank measure can diverge arbitrarily, so reporting both is not redundancy but necessity.

We note the methodological point, since it is cheap to adopt: every inequality we considered for this section was first subjected to randomised counterexample search. That step cost minutes and removed a false claim from the paper.

A.4 The agreement model: sensitivity and failure modes

Sensitivity to the option count K . MATH is free-form, so K in Eq. (1) is not naturally defined. The recovered top mass is nonetheless stable: fitting the 32B MATH unanimity rates under $K \in \{4, 10, 30, 100\}$ gives $m_F \in [0.576, 0.585]$ and $m_T = 0.919$ throughout, so the reported separation $m_T - m_F$ varies only between $+0.334$ and $+0.342$. The reason is that the unanimity probability is dominated by the m^3 term whenever m is well above $1/K$; the uniform tail contributes $(K-1)r^3 = (1-m)^3/(K-1)^2$, which is negligible for $K \geq 4$.

Where the model is wrong. The model reproduces the AUC of agreement to within 0.006 in all three domains (Table 7) but does *not* reproduce the full pattern distribution. On MATH false consensus it predicts a 2–1 rate of 0.444 against an observed 0.350, and an all-distinct rate of 0.356 against 0.450: the model puts too little mass on complete disagreement. This is a direct consequence of the uniform-tail assumption, which spreads the non-preferred mass evenly instead of concentrating it on a few plausible distractors. The AUC prediction is robust to this error because AUC is driven mainly by the unanimity contrast between groups, which the single parameter does capture. We therefore restrict the claim to what is tested: top mass predicts the *discriminative power* of agreement, not the shape of the agreement distribution.

Independence. Eq. (1) treats the three agents as i.i.d. draws. In our protocol the three round-1 responses are independent samples at the same temperature from the same model, so this is exact for round 1, which is the round r_1 -agree is computed on. It would *not* be valid for round 2, where agents see each other’s answers; we do not apply the model there.

A.5 Confidence: the threshold-dependent classification

Table 2 audits answer-agreement only. This appendix holds the corresponding confidence numbers and the classification we would otherwise have put beside them, together with the reason neither belongs in the audit table.

Table 8 applies the same $|AUC| \geq 0.62$ survival rule to confidence. Read literally it gives a non-monotone fail/pass/fail/pass sequence across 3B/7B/14B/32B. We do not interpret that threshold-sensitive sequence as a scaling law.

We do not report this as an audit outcome for two reasons. First, it is not threshold-robust: the pattern holds only for $\tau \in (0.595, 0.642]$, a window of width 0.047 that happens to contain our 0.62 (`thresh_sensitivity.py`). At $\tau = 0.58$ confidence would also “pass” at 14B, and the four-scale sequence would change. The answer-agreement result has no such fragility: it survives for any $\tau \leq 0.642$, so lowering the threshold cannot break it. Second, a pass/fail label placed beside answer-agreement invites a comparison the evidence does not support, because the two are not being certified to the same standard.

What we do claim about confidence is stated in Section 5 in terms of estimated quantities and their intervals, plus the held-out threshold performance of `thresh_perf.py`, neither of which requires a survival threshold at all.

Table 8: Exploratory: mean confidence under the same gates. Not an audit outcome—see the two objections below. Columns match Table 2: within-Level stratified $|AUC|$, the same after removing the independent Mathstral difficulty and official Level, collider amplification, and the pooled mean separation gap = $\bar{c}_T - \bar{c}_F$. The *would-be verdict* column is the mechanical application of the $|AUC| \geq 0.62$ rule to the `−ext−lvl` column and is shown only so that the threshold dependence discussed below is checkable. No max-statistic adjusted permutation p -value is reported: the within-Level permutation test asks which of the eight candidate signals wins, and confidence never does.

Backbone	within	−ext−lvl	amp	gap	would-be verdict
Qwen2.5-3B	0.512	0.504	−0.085	+0.014	fail
Qwen2.5-7B	0.751	0.649	−0.037	+0.049	pass
Qwen2.5-14B	0.694	0.590	−0.011	+0.024	fail
Qwen2.5-32B	0.791	0.702	−0.022	+0.057	pass

A.6 Reproduction

Propositions and their numerical checks: `theory_32b.py` (invariance checks, counterexample search, over-identification test) and `theory_32b.v2.py` (standardised-gap decomposition, bootstrap, label-shuffle controls). Because the converged item sets differ across scales, between-scale comparisons resample each scale’s items independently rather than pairing them; within-scale comparisons of two signals on the same items (e.g. agreement vs. confidence on MMLU) use a genuinely paired bootstrap that scores both signals on the same resampled items. The four-scale bootstrap of Table 1 is `invu_significance.py` (pooled) and `invu4_within.py` (within-Level); the cross-domain power check is `_arc4pt.py`, and the threshold analyses are `thresh_sensitivity.py` and `thresh_perf.py`.

B Execution and artifact record

Most reported intervals are item-resampling intervals conditional on one archived output trace (Section 2); no generation-level confidence interval is claimed. The P0 configurations additionally have one independently generated repeat each. Its paired item-bootstrap intervals condition on the two archived traces and likewise are not generation-level intervals. Runs used Ollama 0.33.3 on Windows with one NVIDIA RTX 3070 (8GB). The local weight blobs used by the P0 runs were: Qwen2.5-3B-Instruct, sha256:5ee4f07cdb9beadbbb293e85803c569b01bd37ed059d2715faa7bb405f31caa6; Qwen2.5-7B-Instruct, sha256:2bada8a7450677000f678be90653b85d364de7db25eb5ea54136ada5f3933730; Qwen2.5-14B-Instruct, sha256:2049f5674b1e92b4464e5729975c9689fcfbf0b0e4443ccf10b5339f370f9a54; Llama-3.1-8B, sha256:667b0c1932bc6ffc593ed1d03f895bf2dc8dc6df21db3042284a6f4416b06a29; Mistral-7B-Instruct, sha256:f5074b1221da0f5a2910d33b642efa5b9eb58cfdddca1c79e16d7ad28aa2b31f; and Mathstral-7B, sha256:9bf04412c5e222bf940d474c4b38c4acc7e8bb8964fb9f41a5c7e8c9388fbd20.

The MATH protocol uses temperature 0.7 and a 1536-token cap. The P0 MMLU holdout uses the same temperature, three samples and two rounds, but a short answer-and-confidence-only prompt with a 128-token cap. Item order is fixed; the backend sampling stream is not. The 200-item holdout manifest records no overlap with the earlier pool and SHA-256 af4f37816da05e36ba80b8b92c3f8e5296b66d9a63f999ccf13acce849f2a798. The final local P0 artifact archive has SHA-256 4ab9a40d02cd4e9b6274e53f3bd09ab53b9dd14f840fa916b4c61b92eae02332.

For the independent repeat, item order used seed 20260911; this seed orders items only and does not seed Ollama sampling. Fresh JSONL logs were collected for Qwen2.5-3B/7B/14B on MATH and

Qwen2.5-3B/7B/14B, Llama-3.1-8B, and Mistral-7B on the P0 MMLU holdout. The usable-record counts were 225/235/238 for MATH and 200 for each MMLU cell; each file contains that many distinct questions. The analysis outputs are `P0_REPEAT_RESULTS.md` and `P0_REPEAT_RESULTS.json`. The repeat archive `P0_REPEAT_FINAL_ARTIFACTS.zip` has SHA-256 `739f7d72532d70cecad4aa13acf6b2aeeb26545e892e5b30beff947cb5db406c`.